

# Development of a technical workflow for a parameter-driven generation of BIM models for standardized reinforced concrete bridges according to the OpenBIM approach

## 1. Parameter schema

| Category       | Parameter               | Symbol     | Typical Value / Range | Source / Reference | Notes (RAB-ING relevance)      |
|----------------|-------------------------|------------|-----------------------|--------------------|--------------------------------|
| Road Geometry  | Road class              | –          | RQ 11                 | FGSV – RAL 2012    | Primary input parameter        |
|                | Total road width        | B          | 11.00 m               | RAL 2012           | Governing width                |
|                | Lane width              | b_lane     | 3.50 m                | RAL 2012           | Fixed for RQ 11                |
|                | Shoulder (Bankett)      | b_shoulder | 1.50 m                | RAL 2012           | Safety + recovery              |
|                | Edge strip              | b_edge     | 0.50 m                | RAL 2012           | Marking + drainage             |
|                | Cross slope             | q          | 2.5%                  | RAL 2012           | Drainage requirement           |
| Deck Geometry  | Total deck width        | B_total    | 12.5 – 13.5 m         | BAST – RiZ-ING     | Includes edge beams            |
|                | Edge beam width (Kappe) | b_kappe    | 0.50 – 0.75 m         | RiZ-ING            | Mandatory durability element   |
|                | Cantilever length       | l_c        | 1.5 – 2.5 m           | RiZ-ING / RAB-ING  | Early design parameter         |
|                | Structural depth        | h          | L/15 – L/25           | DIN EN 1992-2      | First sizing (RAB-ING stage)   |
| Superstructure | Structural system       | –          | Beam bridge           | RAB-ING-Entwurf    | System selection stage         |
|                | Beam spacing            | s          | 2.0 – 3.0 m           | EC2 / practice     | Derived param                  |
|                | Beam height             | h_beam     | 0.8 – 1.6 m           | EC2                | Span dependent                 |
|                | Beam width              | b_beam     | 0.4 – 0.8 m           | EC2                | Constructability               |
|                | Slab thickness          | t_slab     | 0.20 – 0.35 m         | EC2                | Durability + load distribution |
|                | Number of beams         | n          | 4 – 6                 | Derived            | Function of deck width         |
| Substructure   | Abutment type           | –          | Pile abutment         | RAB-ING            | Selected in concept phase      |
|                | Pile diameter           | d_p        | 0.6 – 1.2 m           | EC7 / ZTV-ING      | Soil dependent                 |
|                | Pile spacing            | s_p        | 2.5 – 4.0 m           | Practice / ZTV-ING | Constructability               |
|                | Pile length             | l_p        | variable              | EC7                | Geotechnical input             |
|                | Pile cap thickness      | h_pc       | 1.2 – 2.0 m           | EC2                | Load transfer                  |
|                | Wing wall angle         | $\alpha$   | 30° – 60°             | RiZ-ING            | Earth retention                |

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|------------------------|--------------------------|----------|-----------------------|---------------------|---------------------------|
| Alignment / Layout     | Span length              | L        | 10 – 40 m             | RAB-ING             | Primary design driver     |
|                        | Number of spans          | n_span   | 1 – 3                 | RAB-ING             | Economic optimization     |
|                        | Skew angle               | $\theta$ | 0° – 45°              | RAB-ING             | Affects geometry          |
|                        | Bridge width class       | –        | derived               | RAL + RiZ-ING       | Combined logic            |
| Loads & Design         | Traffic load model       | LM1      | standard              | EC1 (DIN EN 1991-2) | Required for sizing       |
|                        | Load class               | –        | e.g. BK 60            | German NA           | RAB-ING selection         |
|                        | Safety class             | –        | consequence class     | EC0                 | Design stage              |
| Equipment              | Vehicle restraint system | –        | H2 / N2               | EN 1317 / RiZ-ING   | Road safety               |
|                        | Railing                  | –        | standard type         | RiZ-ING             | Pedestrian safety         |
|                        | Drainage slope           | i        | $\geq 2.5\%$          | RAL / RiZ-ING       | Mandatory                 |
|                        | Drainage system          | –        | scupper / pipe        | RiZ-ING             | Design choice             |
|                        | Expansion joint          | –        | if L > 30 m           | RiZ-ING             | Movement control          |
| Durability & Execution | Exposure class           | –        | XC4, XD3              | EC2                 | Environmental             |
|                        | Concrete class           | –        | C30/37 – C50/60       | EC2                 | Strength                  |
|                        | Cover depth              | c_nom    | 35 – 60 mm            | EC2 NA              | Durability                |
|                        | Waterproofing            | –        | required              | ZTV-ING             | Deck protection           |